CERTIFICATE OF CONFORMITY

RAE No: *[RAE No.]*

Name and address of Applicant: All Trades Trailer

Project Name: *[Project name]*

Product Classification: Trailer Certification

Certification was carried within following scope

Australian Standards:

1) **ADR 62/02**
2) **AS4100-1998**

The assessment process has been carried out in accordance with the individual rules and conditions agreed with the applicant.

Valid till: 5 Years From issue date (Yearly Inspection required)

Revision	Date	Revision Details	Prepared By
R0	*[Revision date]*	Initial Release	*[Prepared by (initials)]*

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ATT

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Structural Certification

The trailer described in this report was analyzed for structural integrity as required by ADR 62/02. Professional engineers of Composite Structures Australia certify that the design can safely withstand an ATM of *[ATM]*kg. Check for static stresses and fatigue life as per ADR 62/02 and Australian Standard AS4100-1998 (2016).

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Summary of Compliance

No.	Standard	Compliance requirement	Conformity
1	ADR 62/02 - 14.2.1.1.1 Static Design Load	Longitudinal Tension and Compression For trailers up to 23.5 tonnes 'ATM', (N) 1.5 x 9.81 x 'ATM' (kg)	Full Compliance
2	ADR 62/02 - 14.2.1.2.1 Static Design Load	Transverse Thrust For trailers up to 4.5 tonnes 'ATM', (N) 0.5 x 9.81 x 'ATM' (kg).	Full Compliance
3	ADR 62/02 - 14.2.1.3.1 Static Design Load	Vertical Tension and Compression For trailers up to 4.5 tonnes 'ATM', (N) 0.5 x 9.81 x 'ATM' (kg).	Full Compliance
4	ADR 62/02 - 14.2.2 Static Design Load	'Drawbars' - Dynamic Testing 'Drawbars' for other trailers, a horizontal oscillating force of (kN) +/- 0.6 x 'Coupling' 'D-value' (kN) in the longitudinal direction concurrently for at least 2 million cycles	Full Compliance
5	AS4100-1998-R2016	The structural strength of the trailer apart from ADR62/02	Full Compliance
6	ADR 62/02 - 14.4	'Drawbar' Safety Chain Attachment Points Longitudinal tension (N) = 9.81 x 'ATM' (kg); and vertical tension (N) = 0.5 x 9.81 x 'ATM' (kg).	Full Compliance
7	AS4100-1998-R2016	Coupling attachment points - Coupling plate bolt holes strength	Full Compliance

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Background

Structural analysis and certification services are required for a new build chassis and internal carrying member's design and analysis of a trailer. The trailer requires certification as a Dual Axle trailer type *[ATM]*kg ATM.

3D isometric images of an actual trailer from Autodesk Inventor software identify this product.

[IMAGE: 3D Isometric Model in Autodesk Inventor - Front]

3D Isometric Model in Autodesk Inventor - Front

[IMAGE: 3D Isometric Model in Autodesk Inventor - Back]

3D Isometric Model in Autodesk Inventor - Back

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The geometry of the Trailer

ATT has provided the required 3D geometric data that was required for the structural analysis of the trailer. Shown following are the dimensions of the major structural members of the trailer which was used for the analysis.

Overall length	Overall width	Coupling to suspension
[Overall length] mm	*[Overall width]* mm	*[Distance coupling to suspension]* mm

[IMAGE: 3D Isometric Model]

3D Isometric Model

[IMAGE: Top View showing the length and width of the trailer]

Top View showing the length and width of the trailer

[IMAGE: Side View showing the distance of suspension from coupling]

Side View showing the distance of suspension from coupling

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Requirements

The following requirements apply for the trailer structural compliance as per ADR 62/02,

1. ADR 62/02- Clause 14.2.1.1.1: Longitudinal Tension and Compression For Trailers up to 23.5 tonnes 'ATM' other than 'Converter Dollies', (N) - 1.5 x 9.81 x 'ATM' (kg)
2. ADR 62/02- Clause 14.2.1.2.1: Transverse Thrust For trailers up to 4.5 tonnes 'ATM', (N) 0.5 x 9.81 x 'ATM' (kg)
3. ADR 62/02- Clause 14.2.1.3.1: Vertical Tension and Compression For trailers up to 4.5 tonnes 'ATM', (N) 0.5 x 9.81 x 'ATM' (kg)
4. ADR 62/02- Clause 14.2.2.1: 'Drawbars'- Dynamic Testing 'Drawbars' for other trailers, a horizontal oscillating force of (kN) +/- 0.6 x 'Coupling' 'D-value' (kN) in the longitudinal direction concurrently for at least 2 million cycles.
5. AS 4100-1998-R2016: Structural strength of the trailer not covered in the ADR 62/02 for maximum load condition
6. ADR 62/02- Clause 14.4. 'Drawbars' - Safety Chain Attachment Points
   14.4.1. Any safety chain attachment point affixing a safety chain to a 'Drawbar' must be located as near as practicable to the 'Coupling'. Where two points of attachment are required, they must be mounted one on either side of the centerline of the 'Drawbar'.
   14.4.2. Any safety chain attachment point affixing a safety chain to a 'Drawbar' must withstand at least the following forces, separately applied, without incurring either any residual deformation that would interfere or degrade the function of the assembly or any breaks, cracks or separation of the components. The static forces must be maintained for not less than 10 seconds:
   14.4.2.1. Longitudinal tension (N) = 9.81 x 'ATM' (kg); and
   14.4.2.2. vertical tension (N) = 0.5 x 9.81 x 'ATM' (kg).
7. Coupling mount check

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Analysis- Pre-Processing data

The trailer is analyzed using the finite element analysis technique. The software used for analysis is ANSYS Workbench which is used in the automotive industry for over a decade. 3D elements have been used in modelling as these are the most appropriate for static structural analysis.

Material

The material used for the analysis with the following mechanical properties is given in the below table.

Material	Tensile Strength (MPa)	Yield Strength (MPa)	Young's Modulus (GPa)	Poisson's Ratio	Density (kg/m^3)

Meshing

The fine mesh was created with the number of nodes as *[Mesh nodes]* and the number of elements as *[Mesh elements]*. Shown below are some images of the finite element model that was used for the analysis.

[IMAGE: Meshed model]

Meshed model

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ANALYSIS 1 (ADR 62/02- 14.2.1.1.1) - Longitudinal Tension and Compression

Trailer restraint and the force applied as shown in the below image. The applied force is *[1.5 x 9.81 x ATM]* N as per clause 14.2.1.1.1 of ADR 62/02 i.e., 1.5 x 9.81 x *[ATM]* kg, applied on the coupling holes in the longitudinal direction. All hanger holes faces are fixed in all DOF.

[IMAGE: Load & constraints applied on the trailer]

Load & constraints applied on the trailer

[IMAGE: Stress Distribution Plot for Analysis 1]

Stress Distribution Plot for Analysis 1

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[IMAGE: Stress Distribution Plot with the probe]

Stress Distribution Plot with the probe

As per the result shown in the stress distribution plot, neglecting local stresses, the max stress of *[stress]* MPa was found on the Drawbar, which is compared with the yield strength of material *[Material]* which is *[Yield strength]* MPa.

[IMAGE: Displacement Plot for Analysis 1]

Displacement Plot for Analysis 1

As shown above, the following is the deformation of the structure under the Analysis 1 loading condition. The maximum deformation is *[deformation]* mm.

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ANALYSIS 2 (ADR 62/02- 14.2.1.2.1) - Transverse Thrust

Trailer restraint and the force applied as shown in the below image. The applied force is *[0.5 x 9.81 x ATM]* N as per clause 14.2.1.2.1 of ADR 62/02 i.e., 0.5 x 9.81 x *[ATM]* kg, applied on the coupling holes in the transverse direction. All hanger holes faces are fixed in all DOF.

[IMAGE: Load & constraints applied on the trailer]

Load & constraints applied on the trailer

[IMAGE: Stress Distribution Plot for Analysis 2]

Stress Distribution Plot for Analysis 2

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[IMAGE: Stress Distribution Plot with the probe]

Stress Distribution Plot with the probe

As per the result shown in the stress distribution plot, neglecting local stresses, the max stress of *[stress]* MPa was found on the Hanger, which is compared with the yield strength of material *[Material]* which is *[Yield strength]* MPa.

[IMAGE: Displacement Plot for Analysis 2]

Displacement Plot for Analysis 2

As shown above, the following is the deformation of the structure under the Analysis 2 loading condition. The maximum deformation is *[deformation]* mm.

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ANALYSIS 3 (ADR 62/02- 14.2.1.3.1) - Vertical Tension and Compression

Trailer restraint and the force applied as shown in the below image. The applied force is *[0.5 x 9.81 x ATM]* N as per clause 14.2.1.3.1 of ADR 62/02 i.e., 0.5 x 9.81 x *[ATM]* kg, applied on the coupling holes in the vertical direction (with an equal downward force on the chassis). All hanger holes faces are fixed in all DOF.

[IMAGE: Load & constraints applied on the trailer]

Load & constraints applied on the trailer

[IMAGE: Stress Distribution Plot for Analysis 3]

Stress Distribution Plot for Analysis 3

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[IMAGE: Stress Distribution Plot with the probe]

Stress Distribution Plot with the probe

As per the result shown in the stress distribution plot, neglecting local stresses, the max stress of *[stress]* MPa was found on the Chassis, which is compared with the yield strength of material *[Material]* which is *[Yield strength]* MPa.

[IMAGE: Displacement Plot for Analysis 3]

Displacement Plot for Analysis 3

As shown above, the following is the deformation of the structure under the Analysis 3 loading condition. The maximum deformation is *[deformation]* mm.

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ANALYSIS 4 (ADR 62/02- 14.2.2.1)- Drawbar Dynamic Test

The D-value for the trailer is *[D-value]* kN. The force applied is 0.6 x Coupling D-value = 0.6 x *[D-value]* x 10^3 = *[force]* N.

D-value for this trailer is calculated as shown:
D_C = g . (T x C)/(T + C) (from AS2213.4:2008- 5.1)
Here, T = maximum design total mass in tonnes of the towing vehicle = 2 t
C = mass in tonnes transmitted to the ground by the axle of the centre-axle trailer loaded to its maximum design total mass = *[ATM/1000]* t
Therefore, D_C = 9.81 x (2 x *[C]*)/(2 + *[C]*) = *[D-value]* kN

[IMAGE: Load & constraints applied on the trailer]

Load & constraints applied on the trailer

[IMAGE: Stress Distribution Plot for Analysis 4]

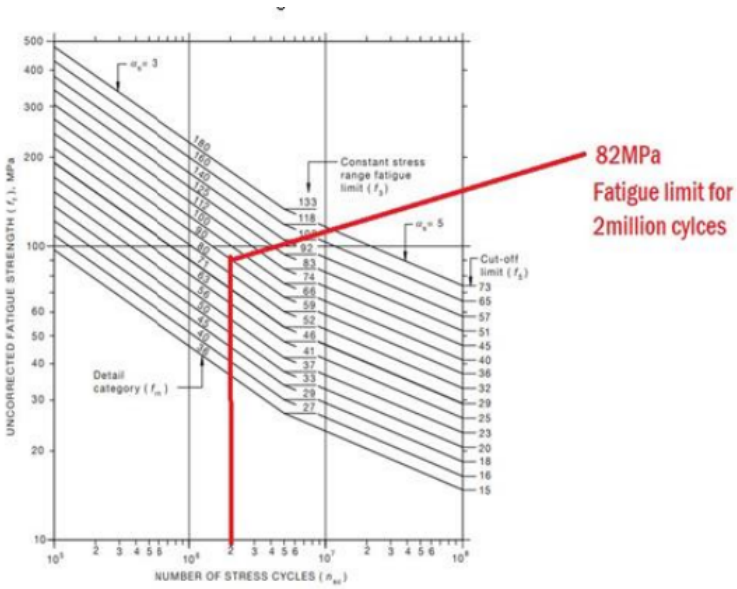
Stress Distribution Plot for Analysis 4

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[IMAGE: Stress Distribution Plot with the probe]

Stress Distribution Plot with the probe

As per result, neglecting local stress the max stress induced in the drawbar of the trailer is *[stress]* MPa, which is lower than the allowable yield limit of *[Yield strength]* MPa. The fatigue limit of 2 million cycles for *[Material]* material as per AS4100-1998, clause 11.6, is 82 MPa. *[verdict]*



Fatigue Criteria as stated in AS4100-1998

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[IMAGE: Displacement Plot for Analysis 4]

Displacement Plot for Analysis 4

As shown above, the following is the deformation of the structure under the Analysis 4 loading condition. The maximum deformation is *[deformation]* mm.

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ANALYSIS 5 (AS4100-1998 2016) - Structural strength of the trailer

For *[ATM]*kg ATM was used for this load case in the chassis. The structural members of the trailer are analyzed with 1G vertical load & 1G Horizontal load applied simultaneously to check the structural integrity of the trailer as per AS 4100-1998.

As shown below following is the distributed loading applied to the trailer and the restraints. This loading is the worst-case scenario in practical application. 1G vertical load = *[load]* N; total forward & downward force (resultant of 1G vertical and 1G horizontal) = *[load]* N; braking force (0.8 x 9.81 x ATM) = *[load]* N.

[IMAGE: Force acting on Chassis top surface & Spring Hangers]

Force acting on Chassis top surface & Spring Hangers

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[IMAGE: Remote Displacement at Coupling Mount holes and Spring Hanger Bracket]

Remote Displacement at Coupling Mount holes and Spring Hanger Bracket

[IMAGE: Reaction forces at Coupling and Spring Bracket respectively]

Reaction forces at Coupling and Spring Bracket respectively

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[IMAGE: Stress Distribution Plot for Analysis 5]

Stress Distribution Plot for Analysis 5

[IMAGE: Stress Distribution Plot with the probe]

Stress Distribution Plot with the probe

As per the result shown in the stress distribution plot, neglecting local stresses, the max stress of *[stress]* MPa was found on the Chassis, which is compared with the yield strength of material *[Material]* which is *[Yield strength]* MPa.

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[IMAGE: Displacement Plot for Analysis 5]

Displacement Plot for Analysis 5

As shown above, the following is the deformation of the structure under the Analysis 5 loading condition. The maximum deformation is *[deformation]* mm.

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ANALYSIS 6 (ADR 62/02- 14.4) - 'Drawbar' Safety Chain Attachment Points

The chain will be fitted on either side of the center line of the drawbar, which satisfies ADR62/02 - 14.4.1.

The longitudinal tension of 9.81 x 'ATM' kg is *[force]* N (vertical tension 0.5 x 9.81 x 'ATM' = *[force]* N). As 2 bolt(s) will be supporting the safety chain, the force per bolt is *[force]* N. The plate thickness is 10mm.

The compressive (bearing) stress induced per bolt = longitudinal force per bolt / (thickness x hole clearance) = *[force]* / (10 x 15) = *[stress]* MPa

[verdict]

[IMAGE: Safety chain mount]

Safety chain mount

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ANALYSIS 7 - Coupling

The pulling force at 1.5G (1.5 x 9.81 x 'ATM' kg) is *[force]* N. As there are 4 bolts, this force per bolt is *[force]* N. The thickness of the coupling plate is 6 mm and the bolt size is M14.

The compressive stress induced per bolt = *[force]* / (Plate thickness x Hole clearance) = *[force]* / (6 x 15) = *[stress]* MPa

[verdict]

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CONCLUSION

The trailer fully complies with ADR 62/02 and AS 4100-1998.

The analysis conducted assumes a new trailer with no material degradation and material are homogenous. Exposure to environmental effects in service may significantly reduce the material's strength capacity to support the full ATM of the trailer. Care must be taken as the trailer will not be able to perform to its full capacity in degraded (e.g., corroded) conditions and an inspection for safety reassessment should be organized by the equipment owner.